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Course Data Analytics Science

Research Assignment 1

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Section A Data Fundamentals

1. A database is a collection of data stored electronically, it is managed by software and it is organised. A database allows data to be stored in tables for easy access, updates and for data to be secured, because modern organisations handle large volumes of data, databases are essential to ensure that the data is secured, accurate and available to multiple users and for making sure the data is protected from unauthorised access.

Examples include banking systems that store customer account information and hospitals that store patient records.

2. A data warehouse is a central repository that stores large volumes of historical data that was collected from multiple source systems. A data warehouse differs from an operational database which was created to handle daily transactions. A data warehouse is designed for analysis, reporting and decision making. Organisations use data warehouses to analyse trends over time and support strategic business decisions.

~~3. A data~~ Examples include a banking data warehouse which integrates data from multiple systems such as customer records, transactions and loans.

Retail Data warehouses (eg Shoprite and Takealot)
This one collects sales, inventory and customer data to analyse purchasing patterns, forecast demand and improve business performance

3. A data lake is a Storage system that holds large amounts of raw data that is in its original format. It can store structured, Semi Structured and Unstructured data. The data lake acts as the initial storage layer before the data is processed and transformed into the Data Warehouse for reporting and analysis. Data stored at a data lake includes text images and logs. Data that does not follow a structure or a strict table format but still ~~has~~ has organisational value is stored here. These can even be social media posts and audio recordings. A data lake preserves data in its original form to maintain history and support downstream processing.

Examples I am aware of includes the Banks data Reservoir. It is used to collect data from customer systems (MOM) and transactional systems. And examples like Transunion and Experian. Google Cloud Storage and Amazon S3 data lake.

4. A data mart is a smaller, focused subset of a data warehouse designed specially for a department or business function. It only contains relevant data, making it faster and easier for users for users such as managers or analysts to access the information they need.

Examples include a finance data mart this is a subset of a data warehouse and it contains financial data such as revenues, expenses and transaction summaries. This is information the finance team and management use to generate reports.

A customer data mart stores customer related information such as profiles, relationships and contact details.

5 Structured data - Is highly organized data this is data that is organized into rows and columns and stored in tables. The data follows a format which makes it easy to store, easy to search and easy to analyze.

examples would be customer tables in a database, employee records and bank account data.

Semi structured data - does not follow a strict table format but contains tags or labels. This is a flexible data format but it is readable by the systems. examples would be JSON files, XML files and API responses.

Unstructured data has no predefined format and includes items such as emails, images and documents. These types of data cannot be easily stored in tables.

Examples extend to audio and social media posts.

6 Data modeling is the process of designing how data is structured, stored and related within a system.

data Modeling is intended to ensure data consistency, clear relationships and efficient storage and retrieval. There are 3 main data models: Conceptual data models, logical data models and Physical data models. These models are used together when designing data systems.

Conceptual = Ideas

Logical = Structure

Physical = Tables

7. Data Mining is the process of analysing large data sets to discover patterns, trends and relationships. Businesses use data mining to improve decisions made, detect fraud, understand customer behaviour and be able to have a futuristic view when making decisions.

Examples include how banks analyse transactions. Looks for unusual patterns, maybe a sudden withdrawal or a sudden large withdrawal in a different location becomes flagged as suspicious and fraud is prevented.

When stores analyse what a customer buys and finds a pattern and it is now able to make product recommendations and increase sales.

8. DBMS is a Database Management System software which is used to create, manage and interact with databases.

Examples include MySQL, PostgreSQL each is used for different applications.

9. ETL stands for Extract, Transform, Load. It is a process used to move data from Source Systems into data Warehouses or analytics platforms. ETL's ensure that data is Cleaned, Standardised and prepared for analysis.

Examples include how Companies extract data from multiple Channels, transform it by removing duplicates and standardising information and load it into a central system to create a single view.

Health care systems that organize patient details or data from different systems and add it to have a unified view of the patient's record for better analysis and health recommendations.

10. Common data storage formats used in analytics include CSV, JSON, Parquet and Avro. Each format has different characteristics and is used depending on the type of data, Performance, needs and use case. JSON is flexible and semi structured.

CSV files are simple and human readable.

Parquet is optimized for analytics performance.

Avro is used for schema based data streaming.

11. Artificial Intelligence refers to the ability of machines or computer systems to perform tasks that normally require human intelligence, such as learning, problem solving, decision making and understanding language. The difference between Narrow AI and General AI.

Narrow AI refers to AI systems that are designed for specific tasks and cannot go beyond that programming function. These are also known as weak AI.

General AI refers to AI systems that can think, learn, and perform any intellectual task that a human can do. These can handle multiple tasks and can even reason.

12 Machine learning is a branch of AI that allows computer systems to learn from data and improve their own performance without being explicitly programmed. Instead of following fixed rules, machine learning systems identify patterns to make predictions or decisions.

Traditional programming relies on predefined rules written by a programmer where rules and data are used to produce an output. Machine learning uses data to learn patterns and create models that generate outputs.

One needs defined programs for it to execute a specified task, and the other one does not rely on direct programming.

Supervised learning uses labels on the data.

Unsupervised learning uses unlabelled data.

Reinforcement learning is based on trial and error.

13 Deep learning is a type of machine learning that uses neural networks to learn from large amounts of data and recognize complex patterns.

Neural networks are brain-inspired systems that process data through layers to produce results.

and improve accuracy over time. Neural Networks work by passing data through different layers these are systems inspired by human brains. Neural Networks improve overtime by adjusting their calculations based on errors

Deep learning is widely used in applications such as image recognition, speech recognition and fraud detection

14 NLP $\rightarrow$ Natural Language Processing is a branch of artificial intelligence that enables computers to understand, interpret and communicate using human language, NLP's allow machines to process text and speech in a meaningful and useful way

Voice assistance when using Siri, Alexa and Google assistance

Chatbots on websites that assist you to use the APP's they answer questions and assist automatically

Text translation like Google translate that are able to translate one language to another

15. ~~LLM~~ LLM $\Rightarrow$ Large Language Model this is an AI model trained on large amounts of text to understand and generate human language. The difference between standard machine learning is that machine learning that is standard does one task while LLMs can do multiple language tasks

Examples would be ChatGPT which answers questions, writes text and Google Gemini will search and summarises

16 Generative AI is a type of AI that creates new content such as text, images or videos based on data it has learned

It works by learning patterns from large amounts of data and then it will generate new content when its given a Prompt

Chat GPT → Generates text, answers questions and helps with writing or coding

DALL-E → Creates images from text description

Google Gemini → Assists with answering questions generating content and information summary

17 Training Datasets this is the data used to train an AI model to learn patterns. The quality of the training data is important because it determines how accurate and reliable the model will be. If the data is biased or incorrect the model may and will produce wrong or unfair data results. Bad training data = A badly trained AI model. If the data is biased or incorrect the model will make wrong predictions, cause real world problems

18 Computer is a branch of AI that enables machines to see and understand images and videos. The machines interpret images by converting them into data and analyzing patterns using algorithms and neural networks. Examples of computer vision includes face recognition systems and self driving cars

19 An API $\Rightarrow$ Is a set of rules that allows different systems to communicate and share data.

A simple analogy that comes to mind is that of a restaurant waiter where the waiter takes a request to the kitchen and returns the response without exposing how the system works. APIs are important because they enable systems to connect, share data and automate processes.

20 An API key is a unique identifier used to authenticate access to an API. Its purpose is to ensure that only authorised users can access the system. API keys should be stored securely and not shared publicly. If an API key is exposed, it can lead to unauthorised access, data breaches and misuse of services.

21 A REST API is an API that follows the principle of representational state transfer (REST) to allow systems to communicate over the internet. The main HTTP methods used in REST APIs are GET $\Rightarrow$ which retrieves data.

POST $\Rightarrow$ creates new data

PUT $\Rightarrow$ updates existing data

DELETE $\Rightarrow$ removes data from a system

22 JSON (JavaScript Object Notation) is a lightweight data format used to store and exchange data using key-value pairs. It is commonly used in APIs because it is easy to read, efficient and widely supported by different systems.

Example `{ "Customer_id": "001", "Name": "kgadi",
"balance": 5000 }`

23 Prompt Engineering is the process of designing clear and specific instructions for AI systems to produce accurate results

It is important because better prompts lead to better and more useful responses.

Examples would be

A Poor Prompt like "Tell me about data" can be improved and it can be "Explain the role of Data Stewardship in banking."

And "Write something about AI" can be improved to something like "Write a short paragraph explaining AI and its uses in everyday life"

24 A map function is used in programming to apply a specific operation to each element in a list and return a new list with results. It is used in data processing to transform data efficiently.

An example in Python the map function can be used to square numbers in a list producing a new list with the results

25 Cloud Computing is the delivery of computing services such as storage, servers, and software over the internet.

The 3 main cloud services are

Infrastructure as a Service (IaaS) - Provides tools for developing applications (e.g. Microsoft Azure, Amazon EC2)

Platform as a Service (PaaS) $\Rightarrow$ Provides tools for developing applications (eg Microsoft Azure App Services)

Software as a Service (SaaS) $\Rightarrow$ provides ready to use applications (eg Microsoft 365)

26 A version Control System is a tool used to track and Manage Changes to files over time
Git $\Rightarrow$ is a widely used version Control System that allows developers and data analysts to Collaborate, track changes and restore previous versions

A Commit $\Rightarrow$ Is a saved Snapshot of Changes
While a branch is a separate version of a Project used for making changes independently

27 A Jupyter Notebook $\Rightarrow$ Is an interactive environment used to write and run code, display results and include explanation in one document.

Jupyter is commonly used with Python
It is popular because it allows them to combine code, helps with visualization and explanations making the data analysis easier and more understandable

28 Python is a high level Programming language widely used in data analytics and artificial intelligence because it is simple, powerful and easy to learn

Its large number of libraries and strong community support is what makes it popular

Common Python libraries includes Pandas for data Analysis

Numpy for Numerical Computation

Matplotlib for data Visualization

Scikit learn for machine learning

29 Data Privacy refers to the protection of personal information and ensuring it is used responsibly.

It is important because it prevents misuse of data and protects individuals rights.

General data Protection Regulation $\Rightarrow$ Is a law that protects personal data and sets rules for how organisations handle it.

GDPR $\Rightarrow$ Applies to countries in the European Union and to any organisation that processes data of EU citizens

30 Data ethics refers to the responsible and fair use of data, ensuring that it is used in a way that protects privacy and promotes fairness

Examples includes

Unethical data use $\Rightarrow$ This includes collecting personal data without consent which violates privacy.

Using biased AI systems which may lead to unfair decisions.

Misusing data to manipulate people which can cause harm and misinformation.

31 Data governance is the process of managing data using policies, rules and standards to ensure it is accurate, secure and properly used

Key components of a good data governance framework includes data quality, data security, clear policies, and standards, defined roles and responsibilities and compliance with regulation.

32 Bias in AI $\Rightarrow$ occurs when an AI model produces unfair or inaccurate results due to biased or unbalanced training data.

This means the model may favour certain group over others, leading to discrimination.

Example would be a hiring system trained on 'biased data' may unfairly prefer certain candidates. The consequences includes unfair decisions, loss of trust and potential issues.

33 Data analysts focus on analysing data and creating reports using tools like Excel and Power BI (These are story tellers)

Data Scientists build predictive models using machine learning and tools like Python (These are prophets)

Data Engineers design and manage data pipelines, using tools such as Spark and SQL.

AI engineers develop advanced AI systems using tools like TensorFlow and PyTorch.

Each role focuses on different parts of the data and AI process.

34 Business Intelligence (BI) is the use of tools and technologies to analyze data and support decision making.

Popular BI tools includes Power BI which

Creates interactive dashboards and reports
Tableau → Helps with visualizing data and identifying trends

These tools help businesses make data driven decisions by turning data into useful insight

35 This assignment helped me connect theoretical concept in data and artificial intelligence to my real work environment. It improved my understanding of how data flows, how its managed and how it is used in within our organisation. I now view banking differently. It was through topics such as data warehouses, data lakes, ETL processes and data modelling, I now better understand how data moves from source system into platforms where it is analyzed and used for decision making. I now better understand my role in ensuring accurate, secure and reliable data for business decision making

36 I found Section C which covers APIs and data exchange the most difficult to understand because it is more technical and focuses on how systems communicate. This is a new topic in my field, I could not reference it to anything I use daily so I relied on course material, online resources like the YouTube videos didn't get enough time to study all the material and finish the assignment. This is a section I have marked and I have tasked myself to do more research and learn more on the topics.

37 Data and Artificial Intelligence will have a significant impact on my future career and daily life. In my career as a data steward, data is already at a center of my job. I am responsible for ensuring data quality, accuracy, and governance, which directly supports business decisions. As organizations rely more on AI, high quality data will become even more important, meaning my role will continue to grow in importance. So, in all honesty, I am hyped up, that is why I even pushed myself to write and finish this whole assignment word by word using Pen and Paper.

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